Analysing Data – Assignment 2 – Lise Feringa

What features does a story need to have to be perceived as a well-written story? The file *data_full_story.csv* contains data derived from comments from the *r/ChangeMyView* subreddit. This online community is designed for structured discussions where users present opinions and invite others to challenge those opinions.

The dependent variable within the dataset is the story strength (story_score). It represents the overall strength or quality of the story in a post. This variable ranges from 0 to 1. This main variable is based on several independent variables that make up the story score altogether. These independent variables (or: features) range from 1 to 5, where 1 represents a weak presence of the feature, and 5 represents a very strong presence. These independent variables are *agency_score* (the presence of actors performing actions), *event_score* (the presence of sequences of events), *world_score* (the envioronment), *curiosity_score* (how much the text stimulates the reader by leaving questions), *surprise_score* (unexpected developments) and *suspense_score* (the level of tension). My task for this assignment is to figure out which variable is the most predictable for the eventual outcome of the story score.

I will begin with plotting scatterplots of the data. Scatterplots tell how strongly each feature might predict the dependent variable *story_score*. The x-cordinate represents the the value of the independent variable (e.g., *agency_score*), and the y-cordinate represents the story_score for the same story. So, if a dot is at (3, 0.8), that post has an agency score of 3 and a story score of 0.8. The scatterplots can be found in Figure 1.

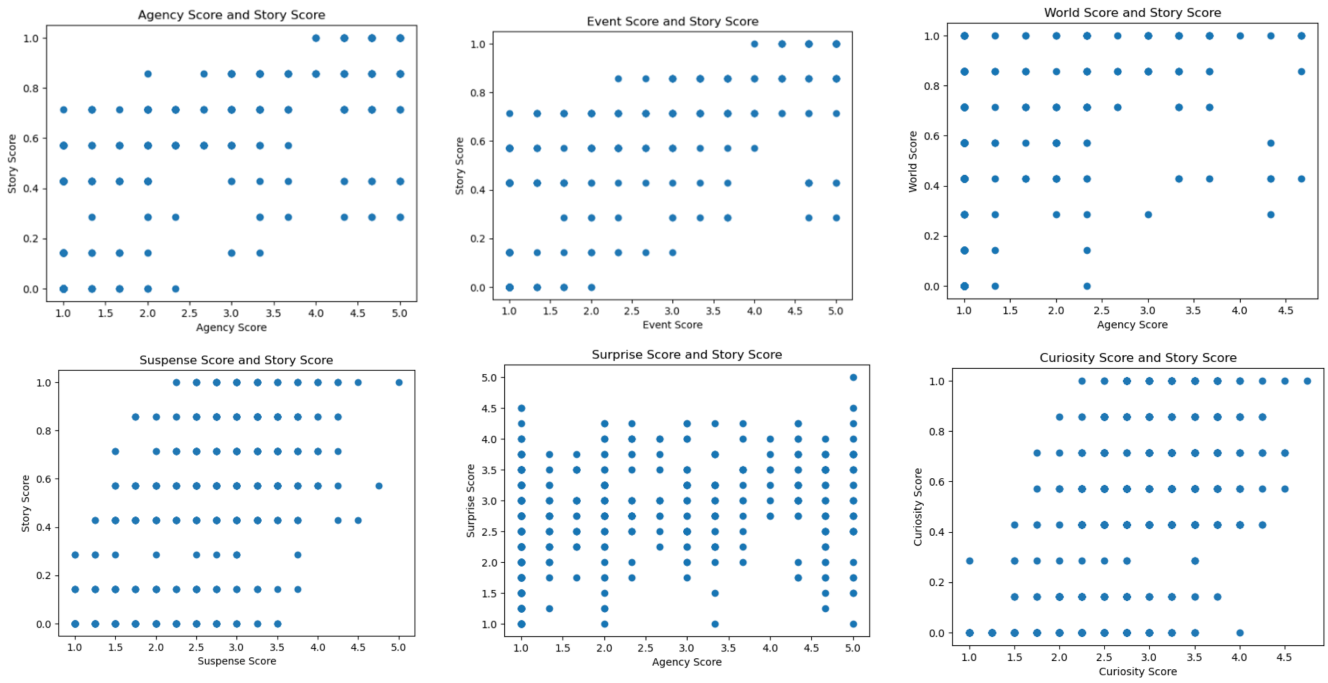

Figure 1. Scatterplots of different features in relation to the story score

Looking at Figure 1, I can make an informed guess about which variables might serve as predictors for the overall story score. *Event_score* and *agency_score* show the most consistent upward trends: as their values increase from left to right, the story_score generally increases, even though a few outliers exist. This suggests that posts with stronger event sequencing and more clearly defined actors tend to have higher story strength, making these two features the most promising weak predictors. Suspense and curiosity also display a slight positive trend,

but the points are more widely scattered, which reduces predictive confidence. In contrast, world score and surprise score show almost no linear pattern.

Next, I will perform simple linear regression. Simple linear regression is a statistical method to model the relationship between two variables. We can use this method to visualize the relationship between two variables by fitting a straight line that predicts the dependent variable (*story_score*) based on the independent variable's values, showing both the direction and strength of the association. The slope of the line indicates how strongly the independent variable affects the dependent variable. A positive slope means that the story_score tends to increase, while a negative slope would indicate a decrease. The magnitude of the slope tells us how much the story_score is expected to change for a one-unit change in the feature.
The intercept represents the predicted value of story_score when the independent variable equals zero.

The results of the plots for each feature can be found in Figure 2.

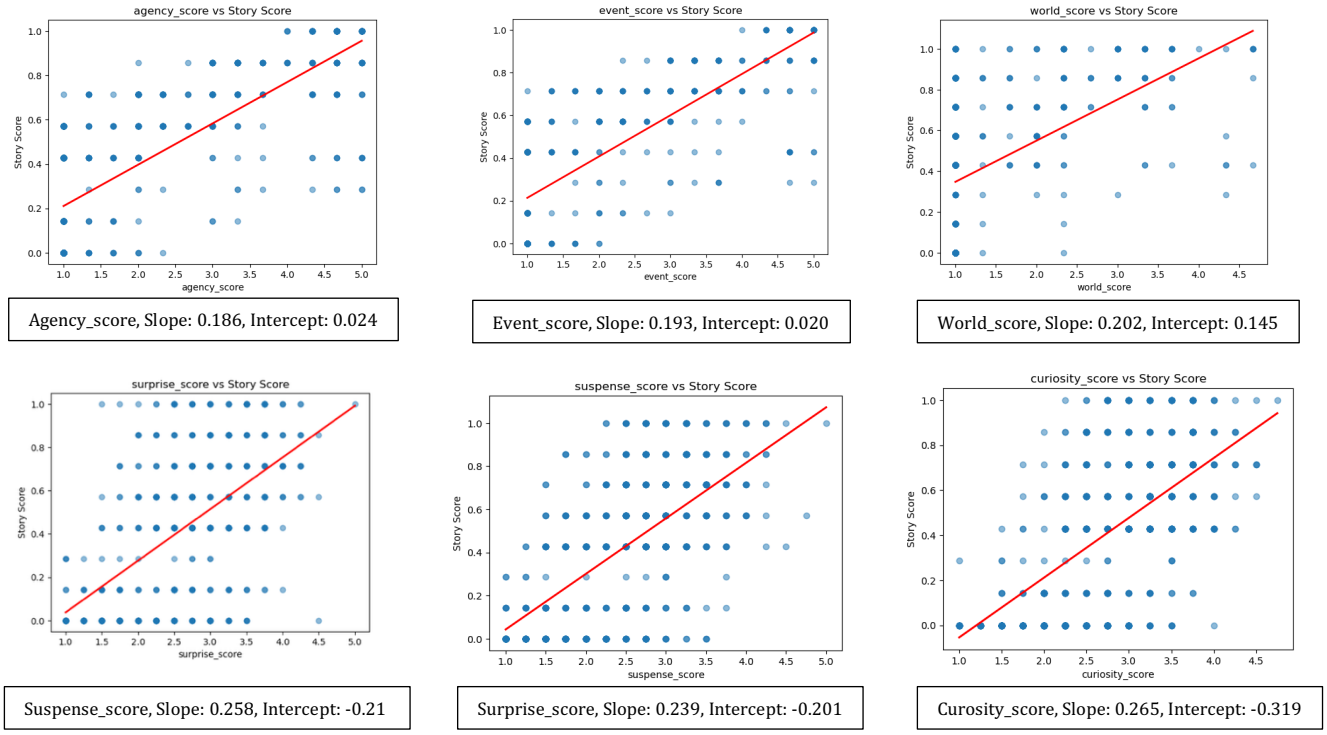

Figure 2. Simple linear regression for each independent variable

All six features have a positive relationship with *story_score*, but the strength of that relationship varies. *Curiosity_score* and *suspense_score* have the steepest slopes, indicating that increases in these features are associated with larger increases in story strength. They have, on the other hand, a lot of outliers. *Event_score* and *agency_score* show gentler but more consistent upward trends, making them visually the most reliable weak predictors.

The R^2 score (coefficient of determination) measures how well the independent variable actually explains the variation in the dependent variable. This is important to establish before reaching a conclusion about what variables might be predicative of the story strength. R^2 ranges from 0 to 1. A score of 0 means that the regression line explains none of the variation; the independent variable does not predict the dependent variable. A score of 1 means that the regression line perfectly explains all variation in the dependent variable. In simple terms: it

tells us how much of the change. in *story_score* can be explained by the feature. The R^2 scores can be found in Figure 3.

agency_score: $R^2 = 0.581$
event_score: $R^2 = 0.597$
world_score: $R^2 = 0.204$
curiosity_score: $R^2 = 0.361$
surprise_score: $R^2 = 0.343$
suspense_score: $R^2 = 0.421$

Figure 3. The R^2 scores of each independent variable.

Event_score and *agency_score* have the highest values. They suggest that around 58–60% of the variation in story strength can be explained by these features alone. This is very consistent with the upward trend observed in their scatterplots.

Residuals will tell what the regression line fails to explain. Randomly scattered residuals indicate a good linear fit; systematic patterns or large deviations indicate poor fit or that the feature is a weak predictor. The residuals are plotted in Figure 4.

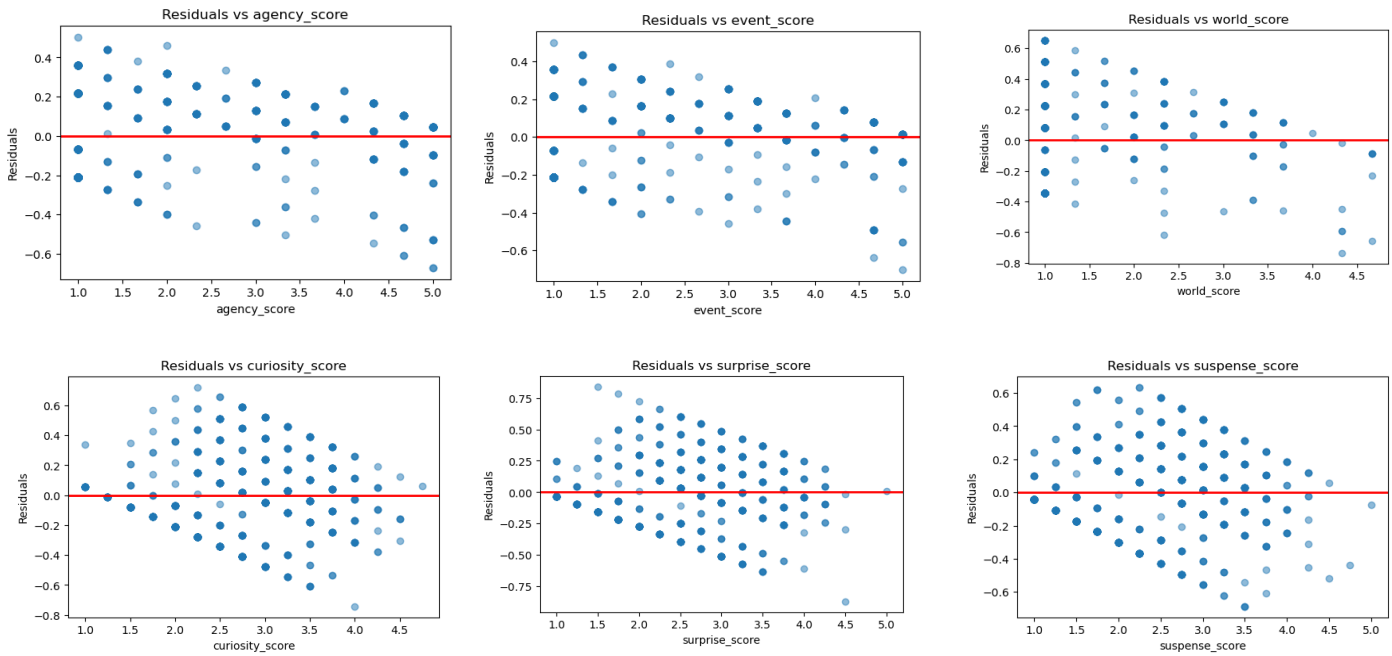

Figure 4. Residuals and all independent variables

According to Figure 4, *agency_score* and *event_score* show the most evenly scattered points around the regression line. This suggests that a linear relationship between these variables and the *story_score* is plausible. The last three variables (*curiosity_score, surprise_score,* and *suspense_score*) show more systematic patterns in their plots. As a result, these variables are weaker predictors, since changes in their values do not consistently correspond to changes in story strength.

Lastly, I will provide a Shapiro-Wilk test on the residuals. The Shapiro–Wilk test checks whether the residuals are normally distributed.

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agency_score: Shapiro-Wilk W=0.943, p=0.000
event_score: Shapiro-Wilk W=0.943, p=0.000
world_score: Shapiro-Wilk W=0.914, p=0.000
curiosity_score: Shapiro-Wilk W=0.989, p=0.000
surprise_score: Shapiro-Wilk W=0.996, p=0.072
suspense_score: Shapiro-Wilk W=0.994, p=0.018
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Figure 5. The results of the Shapiro-Wilk tests for each independent variable.

The *agency_score* and the *event_score* both have p-values < 0.05, even though the R^2 values were relatively high. This indicates that there is some structure or skew in the residuals that the linear model does not capture perfectly. *World_score, curiosity_score* and *Suspense_score* also have a p-value < 0.05: the model captures some trend but leaver unexplained variation.

Surprise_score is the only variable with a p-value > 0.05. This indicates that, although the R^2 is low, the linear model assumptions hold reasonably well for this feature.

In conclusion, *event_score* and *agency_score* are the strongest predictors of story strength. Their positive slopes and moderately high R^2 values indicate that posts with higher event sequencing and stronger agency tend to have higher *story_score.* However, their residuals are not perfectly normal (Shapiro-Wilk $p < 0.05$), suggesting that other unmodeled factors may still influence the story score. *Suspense_score* and *curiosity_score* have moderate slopes and R^2 values, showing a positive relationship with story strength, but the variance explained is smaller. Suprisingly, the residuals of surprise_score are approximately normally distributed (Shapiro-Wilk $p = 0.072$), but the R^2 is low. This indicates that while the linear regression assumptions hold, surprise alone explains little of the variation in story strength, making it a weak predictor. The weakest predictor is *world_score*. Residuals are non-normal, and the scatterplots show little clear trend. This confirms that world building alone contributes minimally to perceived story strength.

Since none of the features fully explain all variation (even the highest $R^2 \approx 0.6$), other factors are likely to (also) influence story strength. These factors may include language clarity (grammar, writing fluency), emotional engagement (empathy, humor) or length. Also elements like pacing or character development may play a role in the perceived good story.